

# Chaehee Park

## AI Research Engineer

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## 1 Research Interests

I specialize in **applied LLM engineering** with a focus on building production-ready AI systems that address real-world challenges. My expertise spans:

- **LLM Engineering & Fine-tuning** — Supervised fine-tuning, prompt engineering, LLMOps; EMS documentation, child safety monitoring, drawing assessment.
- **AI System Architecture** — End-to-end pipelines; RAG, vector DBs, scalable inference infra.
- **AI for Social Impact** — Public safety, child protection, urban policy.

## 2 Professional Experience

Researcher, AI & Big Data Research Division  
Seoul, South Korea

The Seoul Institute  
April 2025 – Present

- **Emergency Response AI System** — Llama-3.1-8B fine-tuned on 90k records; 4.5+ quality; 1.79M+ EMS records processed.
- **Child Safety Monitoring (Watcher Bot)** — LLM-based detection for online CSE/grooming; real-time monitoring pipeline.
- **Policy AI Assistant (RAG)** — RAG-based Q&A, eval dataset, prompt templates for policy proposals.
- **Urban Computer Vision** — YOLO-based BF park facility detection for accessibility compliance.

Full-time AI Researcher  
Seoul, South Korea

Data Science & AI Lab, Sungkyunkwan University  
December 2022 – February 2025

- Multimodal learning & VLMs for psychological assessment; sketch description/summarization.
- YOLOv8 cross-domain generalization; 1,399 scene-level drawing dataset.
- Real-time AI web apps with FastAPI/GraphQL; LLM feedback system deployment.

SKT AI Fellowship (5th Cohort)  
Seoul, South Korea

SK Telecom  
May 2023 – October 2023

- Production-grade **AI copywriting system**; knowledge graph from 3M+ records; LLMOps pipeline.
- A/B testing via MMS; **Best Communicator** award.

Language AI Assistant  
Seoul, South Korea

NLP Center, NC Soft Corporation  
June 2022 – September 2022

- Participated in **Aspect-Based Sentiment Analysis (ABSA)** research; constructed multi-domain sentiment datasets (finance, sports, news).
- Preprocessed and annotated domain-specific corpora; supported development and fine-tuning of **language understanding models**.

## 3 Key Projects

5W1H-Based EMS Log Structuring System

LLM FT · Prompt · Auto Eval

- Llama-3.1-8B on 72k samples with 5W1H prompting; **Claude-based** evaluation (90%+ with 5/5).
- Inference for 1.79M records; PII masking; expert human evaluation protocol.

RAG-Based Policy Knowledge System

Vector DB · Retrieval · QA

- Hybrid retrieval (dense+sparse); optimized chunking; **ChromaDB** vector store.

AI-Driven Art Therapy Assessment Platform

Multimodal LLM · CV · Web

- MLLM-based psychological drawing analysis and education; 5 papers (WWW, CIKM, etc.); real-time web prototype.

## 4 Technical Skills

|             |                                                                                       |
|-------------|---------------------------------------------------------------------------------------|
| LLM & GenAI | Fine-tuning (LoRA/Full), Prompt Eng., LLMOps, Model Eval., GPT-4/Claude, Llama, Gemma |
| ML/DL       | PyTorch, TensorFlow, Transformers, LangChain, YOLO, FastAPI                           |
| Data/Infra  | RAG, ChromaDB/Faiss, MySQL/PostgreSQL, Batch, Docker                                  |
| Programming | Python, C/C++, JavaScript, Linux, Git                                                 |

## 5 Education

M.S., Applied Artificial Intelligence

Advisor: Prof. Jin-Young Han

GPA: 4.25/4.5

Sungkyunkwan University, Seoul

March 2023 – February 2025

B.S., Human-Centered AI & Applied AI (Double Major)

GPA: 4.22/4.5

Sangmyung University, Seoul

March 2019 – February 2023

## 6 Publications

Full list: 6 papers including ACM tier-1 conferences (WWW, CSCW, CIKM) and high-impact journal (TiiS, IF=3.9)

- [1] C. H. Park\*\*, M. G. Yang\*\*, et al. "CheckDAPR: An MLLM-based DAPR Scoring System for Art Therapy" *ACM CIKM*, 2025 (Just accepted).
- [2] M. G. Yang, C. H. Park, et al. "PracticeDAPR: An AI-based Education-Supported System for Art Therapy" *ACM CSCW*, 2025.
- [3] J. W. Kang\*\*, J. W. Kim\*\*, M. G. Yang\*\*, C. H. Park\*\*, et al. "Developing an AI-based Explainable Expert Support System for Art Therapy" *ACM TiiS*, IF=3.9, December 2024.
- [4] J. W. Kang, J. W. Kim, M. G. Yang, C. H. Park, et al. "SceneDAPR: A Scene-Level Free-Hand Drawing Dataset for Web-based Psychological Drawing Assessment" *Proceedings of the ACM Web Conference (WWW)*, May 2024.
- [5] C. H. Park, M. G. Yang, et al. "A Study on Ensemble Model for Predicting Prompts in Images Generated by Diffusion Model" *Korean Institute of Broadcast and Media Engineers Summer Conference*, June 2023.
- [6] E. H. Jung\*\*, C. H. Park\*\*, H. J. Hong\*\*, et al. "On the Deep Generative Models Explaining the Rationale to Emotionally Supportive Conversations" *Korean Society of Emotion and Sensibility Autumn Conference*, October 2022 (Best Paper Award).

## 7 Awards & Honors

- BK Outstanding Student Scholarship, Sungkyunkwan University (2024)
- Best Communicator Award, SKT AI Fellowship 5th Cohort (2023)
- Best Paper Award, Korean Society of Emotion & Sensibility Conference (2022)
- SW Scholarship (2019–2022)

## 8 Teaching Experience

- Undergraduate Research Program (URP), Sungkyunkwan University (Summer 2023)
- Computing Thinking & Game Design, Sangmyung University (Spring 2022)
- Algorithms & Game Contents, Sangmyung University (Fall 2021)